The trigonometric tables give the values, correct to four places of decimals for the angles from 0° to 90° spaced at intervals of 60′. A trigonometric table consists of three parts.

A column on the extreme left which contains degrees from 0° to 90°, followed by ten columns headed by 0′, 6′, 12′, 18′, 24′, 30′, 36′, 42′, 48′ and 54′.

Five columns under the head mean difference has values from 1,2,3,4 and 5.

For angles containing other measures of minutes (that is other than 0′, 6′, 12′, 18′, 24′, 30′, 36′, 42′, 48′ and 54′), the appropriate adjustment is obtained from the mean difference columns.

The mean difference is to be added in the case of sine and tangent while it is to be subtracted in the case of cosine.

Now let us understand the calculation of values of trigonometric angle from the following examples.

Example 6.11 Find the value of $\sin 64°34'$.

Solution

	0′	6′	12′	18′	24′	30′	36′	42′	48′	54′	Mean Difference				
	0.0°	0.1°	0.2°	0.3°	0.4°	0.5°	0.6°	0.7°	0.8°	0.9°	1	2	3	4	5
64°						0.9026								5	

Write $64°34' = 64°30' + 4'$

From the table we have, $\sin 64°30' = 0.9026$

Mean difference for $4' =$ 5 (Mean difference to be added for sine)

$\sin 64°34' = 0.9031$

Example 6.12 Find the value of $\cos 19°59'$

Solution

	0′	6′	12′	18′	24′	30′	36′	42′	48′	54′	Mean Difference				
	0.0°	0.1°	0.2°	0.3°	0.4°	0.5°	0.6°	0.7°	0.8°	0.9°	1	2	3	4	5
19°										0.9403					5

Write $19°59' = 19°54' + 5'$

From the table we have,

$\cos 19°54' = 0.9403$

Mean difference for $5' =$ 5 (Mean difference to be subtracted for cosine)

$\cos 19°59' = 0.9398$

6 Trigonometry.indd 236 02/12/2022 21:48:03